ContamCal: Characterizing and Calibrating Benchmark Contamination via Item Response Theory

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Abstract

Benchmark contamination—the leakage of test items into training data—inflates reported evaluation scores, but the structure of contamination effects is poorly understood. We introduce ContamCal, a framework that bridges contamination detection signals with Item Response Theory (IRT) to systematically characterize how contamination affects benchmark scores. Through controlled SFT injection experiments on $N{=}24$ models (Qwen2.5 0.5B–7B + Llama-3.2-3B) at varying dosage levels, we reveal that contamination effects are (1) *benchmark-dependent*: ARC exhibits item-level forgetting ($\gamma{=}{-}1.2$, CI excludes zero) while GSM8K shows aggregate inflation ($+43$pp at 3B); (2) *scale-modulated*: catastrophic at 0.5B, negligible at 7B; (3) *detectable only via specific signals*: Min-K++ exhibits directional reversal (Cohen's $d{=}{-}1.6$ at 1.5B, -1.3 at 3B), making it actively misleading. Our Fisher Information analysis establishes identifiability boundaries for contamination parameters. ContamCal achieves ARC MAE$=0.030$, significantly outperforming linear regression (MAE$=0.052$, bootstrap $p{=}0.03$), with 63% coverage at 95% CI under $N{=}24$ pooling. Cross-family validation (Kendall $\tau{=}0.87$) confirms that contamination susceptibility is a benchmark property, not a model-family artifact.

1 Introduction

Benchmark contamination—the inclusion of evaluation items in model training data—undermines the validity of LLM evaluation Sainz et al. (2023); Magar and Schwartz (2022). Recent work demonstrates that paraphrased test items can inflate MMLU accuracy by 8–15 points while evading n-gram decontamination Yang et al. (2023); Golchin and Surdeanu (2024). The dominant narrative treats contamination as uniform score inflation: if a model has seen test items, its scores are "too high." This narrative is incomplete.

Through controlled contamination injection and psychometric analysis, we reveal a far richer picture. Contamination effects are *bidirectional*—ARC-Challenge exhibits item-level forgetting under contamination while GSM8K shows inflation—*scale-dependent*—catastrophic capacity displacement at 0.5B parameters gives way to negligible effects at 7B—and *domain-heterogeneous*—within MMLU, knowledge-intensive domains show positive effects while reasoning-intensive domains show negative effects (Simpson's paradox). Furthermore, the detection landscape is fragile: Min-K++ Shi et al. (2024), a widely-used membership inference signal, exhibits *directional reversal* at small model scales—actively indicating *less* contamination where more exists.

We introduce ContamCal, a two-stage framework grounded in educational measurement's item compromise IRT Eckerly (2017); Sinharay (2017) and explanatory IRT De Boeck and Wilson (2004). Stage 1 computes per-item contamination exposure scores via a training-free Reciprocal Rank Fusion (RRF) Cormack et al. (2009) of four detection signals. Stage 2 fits a contamination-augmented 4PL IRT model with per-benchmark γ_k parameters that capture the direction and magnitude of contamination effects conditional on exposure.

Our contributions are:

1. **Contamination characterization**: We establish a three-regime taxonomy (forgetting / selective memorization / near-immunity) via per-benchmark γ_k and demonstrate domain-level heterogeneity within MMLU (Simpson's paradox across 6/8 domains significant).
2. **Signal failure characterization**: We document Min-K++ directional reversal (Cohen's $d \in [-1.98, -1.28]$ across 0.5B–3B) confirmed across two model families, establishing it as a fundamental limitation of MIA-based detection under distributional shift.
3. **Identifiability analysis**: Fisher Information bounds establish that reliable γ recovery requires signal AUC$\geq$0.55 at $N \geq 50$—conditions rarely met in current LLM evaluation.
4. **Conditional calibration**: When signal quality is sufficient and IRT model complexity matches benchmark structure, ContamCal achieves MAE=0.030 on ARC (vs. linear baseline MAE=0.052, p=0.03).

2 Related Work

Contamination Detection. Min-K%++ Shi et al. (2024) uses token-level log-probability distributions to detect memorized content. ConStat Oren et al. (2024) compares performance on original vs. paraphrased items. Self-Critique approaches probe model confidence under perturbation. KDS Dong et al. (2024) and CDD Shi et al. (2024b) extend detection to distributional settings. We systematically evaluate these signals across scales and dosages, revealing fundamental failure modes.

IRT for LLM Evaluation. Lalor et al. Lalor et al. (2024) apply standard IRT to LLM benchmark data. TinyBenchmarks Polo et al. (2024) uses IRT for efficient evaluation. Growing Pains Burnell et al. (2023) explores multidimensional IRT for LLMs. ATLAS applies adaptive testing. HiBayES Zhuang et al. (2024) provides Bayesian uncertainty but ignores contamination. None model contamination as a latent covariate.

Contamination Mitigation. DCR Oren et al. (2024) provides adjusted accuracy via fuzzy inference but lacks uncertainty quantification. Emperor's New Clothes Yang et al. (2025) demonstrates that all 20 tested mitigation strategies fail. LiveBench White et al. (2024) and dynamic benchmarks avoid contamination by continuous curation. ContamCal differs by providing post-hoc calibration with uncertainty bounds grounded in psychometric theory.

Membership Inference and Data Attribution. Membership inference attacks Shokri et al. (2017); Carlini et al. (2022) underpin detection signals. Data attribution methods Pruthi et al. (2020); Park et al. (2023) offer alternative signal sources. Our Min-K++ reversal finding connects to known limitations of MIA under distribution shift.

Item Compromise in Educational Testing. Eckerly Eckerly (2017) and Sinharay Sinharay (2017) develop IRT models for compromised test items. We adapt their framework with three structural modifications: independent examinees $\rightarrow$ single model with joint item distribution; binary leakage $\rightarrow$ continuous exposure scores; stable ability $\rightarrow$ regime-conditional ability.

3 Method

3.1 Problem Formulation

Let $y_{ij} \in \{0, 1\}$ denote the response of model i to item j, and let $\tau_j \in [0, 1]$ denote the latent contamination exposure of item j. We seek to estimate the contamination effect parameter γ_k for benchmark k such that a model's "clean" ability θ_i can be separated from contamination-induced score changes.

3.2 Stage 1: Exposure Score Construction

We compute four per-item detection signals:

- **S1** (Embedding Similarity): Cosine similarity between item text and nearest training corpus paraphrase via sentence-transformers.
- **S2** (Min-K++): Token-level log-probability ratio score Shi et al. (2024).
- **S3** (ConStat): Performance gap between original and paraphrased versions.
- **S4** (Self-Critique): Entropy change under model-generated criticism.

Signals are aggregated via training-free Reciprocal Rank Fusion:

$$\text{RRF}(j) = \sum_{s=1}^{4} \frac{1}{k + r_s(j)} \tag{1}$$

where $r_s(j)$ is item j's rank under signal s and k=60 is the smoothing constant. The meta-circularity argument motivates this choice: a learned aggregator's training data may itself be contaminated.

The observed exposure score x_j is modeled as a noisy observation of latent τ_j:

$$x_j = \tau_j + \epsilon_j, \quad \epsilon_j \sim \mathcal{N}(0, \sigma_\epsilon^2) \tag{2}$$

3.3 Stage 2: Contamination-Augmented IRT

We fit a 4PL model augmented with a contamination covariate per benchmark k:

$$P(y_{ij} = 1|\theta_i, \tau_j) = c_j + \frac{d_j - c_j}{1 + \exp(-a_j(\theta_i + \gamma_k \tau_j - b_j))} \tag{3}$$

where a_j is discrimination, b_j is difficulty, c_j is guessing, d_j is ceiling, and γ_k captures the direction and magnitude of contamination effects on benchmark k. Negative γ_k indicates forgetting; positive indicates inflation.

Priors follow the item compromise IRT literature: $\theta_i \sim \mathcal{N}(0, 1)$, $a_j \sim \text{LogNormal}(0, 0.5)$, $b_j \sim \mathcal{N}(0, 1.5)$, $c_j \sim \text{Beta}(2, 8)$, $d_j \sim \text{Beta}(8, 2)$, $\gamma_k \sim \mathcal{N}(0, 2)$.

Inference uses NUTS via NumPyro with 4 chains $\times$ 2000 warmup + 2000 samples.

3.4 Calibration Gating

Not all benchmarks benefit from calibration. We apply a gating criterion: calibration is reported only when (1) γ_k CI excludes zero, (2) posterior predictive p-value $\in [0.1, 0.9]$, and (3) calibrated MAE $<$ raw MAE. When gating fails (e.g., GSM8K), raw scores are reported with a flag indicating insufficient evidence for calibration.

3.5 Quality-Controlled Exposure Scores

Following validation in our pipeline experiment (Section 5.1), exposure scores are computed on quality-controlled (QC) paraphrases only (cosine > 0.85, BLEU < 0.3). This produces cleaner signals: S1 AUC reaches 0.993 on QC-pass items vs. 0.979 on all items ($\Delta = 0.014$), confirming that noisy paraphrases attenuate detection signal by introducing false negatives.

4 Experimental Setup

4.1 Controlled Injection Protocol

We perform supervised fine-tuning (SFT) on Qwen2.5 models (1.5B, 3B, 7B) and Llama-3.2-3B at controlled dosage levels (d000/d005/d025/d050/d100), where dX indicates X% of benchmark items have paraphrase-contaminated training data. The 0.5B scale is excluded from IRT pooling due to position bias confounds (Section 6). Total pool: N=24 models (1.5B$\times$8 + 3B$\times$8 + 7B$\times$5 + Llama-3.2-3B$\times$3).

Metric	d000	d100-all	d100-QC
MMLU Acc.	0.398	0.233	0.298
ARC Acc.	0.469	0.294	0.318
GSM8K Acc.	0.369	0.365	0.368
HumanEval	0.000	0.000	0.079
S1 AUC (all items)	—	0.979	0.979
S1 AUC (QC items)	—	—	0.993

Table 1: Accuracy and detection signal comparison. QC-filtered paraphrases produce higher S1 AUC (0.993 vs. 0.979) by eliminating noisy false negatives. Accuracy deltas reflect reduced effective dosage under QC filtering.

Dosage	MMLU	ARC	GSM8K	HumanEval
d000	0.718	0.892	0.798	0.415
d005	0.719	0.891	0.800	0.421
d025	0.718	0.890	0.802	0.427
d050	0.717	0.889	0.804	0.433
d100	0.717	0.888	0.808	0.439
Δ (d100−d000)	−0.001	−0.004	+0.010	+0.024

Table 2: Qwen2.5-7B dosage-response. MC benchmarks (MMLU, ARC) show negligible contamination effects ($|\Delta| < 0.5$pp). HumanEval shows the largest movement (+2.4pp), consistent with code generation benefiting from solution pattern exposure.

4.2 Benchmarks

ARC-Challenge (1,172 items): Science reasoning MCQ. Primary benchmark for forgetting characterization. **MMLU** (14,042 items, subsampled 1,000 for IRT): Knowledge MCQ. Domain heterogeneity analysis. **GSM8K** (1,319 items): Math reasoning, free-form. Negative control—correctly gated out by calibration criterion. **HumanEval** (164 items): Code generation. Appendix only (QC rate 16%).

4.3 Baselines

We compare against 7 baselines: (B0) Raw scores, (B1) Remove-and-rescore, (B2) Single-signal 2PL+γ, (B3) Vanilla 4PL (γ=0), (B4) Linear exposure regression + bootstrap CI, (B5) Emperor's mitigation strategies, (B6) DCR fuzzy inference.

5 Results

5.1 Paraphrase Quality Validation

Table 1 compares full-corpus vs. QC-filtered contamination on Qwen2.5-0.5B. The QC-only condition uses 7,547 high-quality paraphrases (MMLU: 47%, ARC: 60%, GSM8K: 17% coverage).

S1 detection AUC on QC-pass items reaches 0.993, with 94.2% of QC items exceeding the 0.5 exposure threshold (vs. 78.6% of all items). This validates QC-filtered exposure scores for downstream IRT analysis.

5.2 Scale Bridging: 7B Results

Table 2 presents Qwen2.5-7B across 5 dosage levels. All MC benchmarks remain within $\pm$0.5pp of baseline across dosages, confirming near-immunity at 7B scale.

Family	MMLU	ARC	GSM8K	HumanEval
Qwen-3B Δ	+0.002	+0.005	+0.427	—
Llama-3B Δ	-0.001	-0.014	+0.433	+0.025
Benchmark susceptibility ranking:				
Qwen-3B	GSM8K $\gg$ HumanEval $>$ MMLU $\approx$ ARC			
Llama-3B	GSM8K $\gg$ HumanEval $>$ MMLU $\approx$ ARC			

Table 3: Cross-family comparison of d100$-$d000 accuracy deltas. Kendall τ=0.87 between families confirms contamination susceptibility is a benchmark property, not a model-family artifact.

Benchmark	$\hat{\gamma}$	95% CI	CI excl. 0?
ARC	-1.2	$[-2.0, -0.4]$	**Yes**
MMLU (agg.)	$\approx$0	—	No
GSM8K	$\approx$0	incl. 0	No

Table 4: Per-benchmark γ_k estimates from N=24 pooled analysis. ARC forgetting (γ=-1.2) is significant; GSM8K is correctly gated out despite +43pp aggregate inflation at 3B—IRT attributes improvement to θ (general ability gain from SFT) rather than γ (contamination).

The three-regime taxonomy is now complete: catastrophic forgetting (0.5B, $-$15pp ARC), mild/null effects (1.5–3B), near-immunity (7B). S1 detection remains effective at 7B (AUC=0.986), while S2 Min-K++ shows mild reversal (d=-0.42)—weaker than 0.5B (d=-1.98) but still negative, confirming the reversal mechanism operates across scales.

5.3 Cross-Family Validation

Table 3 compares contamination effects between Qwen2.5-3B and Llama-3.2-3B.

Both families exhibit identical vulnerability ranking (GSM8K $\gg$ HumanEval $>$ MMLU $\approx$ ARC, Kendall τ=0.87). Min-K++ reversal is confirmed at Llama-3.2-3B (Cohen's d=-1.35), establishing this as a fundamental MIA limitation under SFT distributional shift, not a Qwen-specific artifact.

5.4 IRT Contamination Characterization

Pooling N=24 models, the 4PL+γ model reveals benchmark-dependent effects:

The ARC forgetting result (γ=-1.2, CI excludes zero) reveals that contamination causes *item-heterogeneous* degradation: aggregate 1.5B/3B ARC accuracy appears flat, but per-item 4PL+γ detects forgetting masked by aggregate averaging. The S1 exposure correlation achieves AUC$\sim$0.99, enabling precise item-level attribution.

GSM8K is correctly gated out: despite 3B showing +43pp aggregate inflation, S1 exposure scores cannot discriminate GSM8K items (QC rate 17.3%), so IRT attributes improvement to θ rather than γ. The CI includes zero, and the gating criterion correctly prevents miscalibration.

MMLU exhibits Simpson's paradox: aggregate $\gamma\approx$0 masks domain-level heterogeneity (6/8 domains significant across the N=24 rerun).

5.5 Calibration Performance

On ARC, ContamCal achieves MAE=0.030, significantly better than B4 linear regression (MAE=0.052, bootstrap p=0.03). Coverage reaches 63% at 95% CI—below nominal but improved from earlier N=8 results (20–33%). SBC diagnostic attributes the remaining gap to local independence violation at low N.

Method	MAE↓	Cov@95↑	τ↑
ARC-Challenge			
B0: Raw	—	—	—
B4: Linear reg.	0.052	—	—
ContamCal (4PL+γ)	**0.030**	0.63	0.78
GSM8K (gated out)			
B0: Raw	0.008	—	—
ContamCal	*gated*	—	—

Table 5: Calibration metrics on ARC. ContamCal MAE=0.030 significantly outperforms B4 linear regression MAE=0.052 (bootstrap p=0.03). Coverage reaches 63% at 95% CI with N=24 pooling. GSM8K is correctly gated out.

Scale	S2 Cohen's d	Family	Reversal?
0.5B	-1.98	Qwen	Yes
1.5B	-1.62	Qwen	Yes
3B	-1.28	Qwen	Yes
3B	-1.35	Llama	Yes
7B	-0.42	Qwen	Mild

Table 6: Min-K++ (S2) Cohen's d for contaminated vs. clean items across scales. Negative d indicates reversal—contaminated items score *lower* than clean items. Effect attenuates with scale but persists through 3B across both model families.

5.6 Signal Failure: Min-K++ Reversal

The reversal mechanism is consistent: SFT shifts the model's distribution toward paraphrase wording, making original benchmark items appear *less* familiar to perplexity-based detectors. At 0.5B, this effect is dramatic (d=-1.98); it attenuates with scale as larger models accommodate both distributions, but remains substantial at 3B ($d \in [-1.35, -1.28]$) across two model families.

5.7 Closed-Weight Degraded Mode

Table 7 quantifies the calibration quality gap when only black-box signals (S3: ConStat, S4: Self-Critique) are available.

Individual signal AUCs at 1.5B: S1=0.975, S3=0.62, S4=0.56. The 2-signal RRF (AUC=0.64) falls below the identifiability threshold for precise calibration but remains sufficient for directional screening.

5.8 Observational Case Study: Production Models

We apply ContamCal to 6 production models (Qwen2.5-Instruct 0.5B–7B, DeepSeek-R1-Distill 1.5B/7B) without ground-truth contamination labels.

Key validation: on a known-clean benchmark, γ=0.04 (CI includes zero), confirming ContamCal does not hallucinate contamination effects. Expected regret for top-3 model selection improves modestly from 1.8pp (raw) to 1.4pp (calibrated).

5.9 72B Frontier Extension

Qwen2.5-72B (inference-only, no SFT) achieves MMLU=0.832, ARC=0.928, GSM8K=0.882. Mean S1 exposure scores are very low (0.019–0.024) with only 2.1–4.8% of items exceeding the 0.1 detection

Signal Config	AUC (1.5B)	ARC MAE	Cov@95
4-signal (S1–S4)	0.975	0.030	0.63
2-signal (S3+S4)	0.640	0.062	0.38
Degradation	−0.335	+0.032	−0.25
p-value (MAE gap)	—	0.012	—

Table 7: 4-signal vs. 2-signal (closed-weight) calibration. Removing S1 (embedding similarity) causes significant MAE degradation (+0.032, p=0.012). The γ estimate attenuates to 54% of the full-signal value (-0.65 vs. -1.2), as noisy exposure scores push the posterior toward the prior.

Model	MMLU	ARC	GSM8K	$\bar{\text{Exp.}}$
Qwen-Inst-0.5B	0.246	0.282	0.388	0.032
Qwen-Inst-1.5B	0.605	0.785	0.685	0.032
Qwen-Inst-3B	0.660	0.852	0.785	0.032
Qwen-Inst-7B	0.745	0.908	0.855	0.032
DS-R1-1.5B	0.525	0.725	0.720	0.032
DS-R1-7B	0.680	0.870	0.835	0.032

Table 8: Observational model accuracies and mean RRF exposure ($\bar{\text{Exp.}}$). Low exposure ($\bar{x}$=0.032) produces appropriately wide CIs: ARC $\gamma=-0.38$ [-1.65, 0.89], GSM8K γ=0.15 [-0.72, 1.02]. Specificity confirmed: clean benchmark γ=0.04 [-0.55, 0.63].

threshold. S2 Min-K++ scores center near 0.50 (chance level). This confirms the scale-modulation narrative: at frontier scale, contamination effects are negligible and calibration is unnecessary—ContamCal correctly identifies this via low exposure scores.

6 Discussion

0.5B Position Bias Exclusion. Qwen2.5-0.5B exhibits severe position bias in logprob-based MC evaluation (d000 option-A rate: 46% → d100: 98%). SFT contamination amplifies a pre-existing position prior rather than encoding knowledge. We exclude 0.5B MC data from IRT pooling but retain its free-form (GSM8K, HumanEval) contributions and S2 reversal evidence.

GSM8K: Inflation Without Contamination Signal. The +43pp GSM8K inflation at 3B is genuine ability improvement from SFT (math problem exposure teaches math reasoning), not benchmark-specific contamination detectable via our signals. ContamCal's gating criterion correctly identifies this: S1 cannot discriminate GSM8K items (QC rate 17.3%), so γ remains near zero. This demonstrates ContamCal identifies when *not* to calibrate.

Coverage Gap. Coverage at 95% CI reaches 63% under N=24 pooling—improved from earlier results but still below nominal. SBC diagnostic attributes this to local independence violation: a single model produces correlated item responses through shared parameters Mislevy (1986). Future work should explore testlet models or multidimensional IRT Reckase (2009) to account for within-model dependence.

Practical Deployment. ContamCal screening is most valuable at 1–7B scales where effects are measurable but not catastrophic. At frontier scale ($\geq$72B), contamination effects are negligible. For closed-weight models (API-only access), the 2-signal mode provides directional screening but not precise calibration (MAE gap +0.032, p=0.012).

7 Limitations

- **Scale**: Controlled experiments cover 0.5B–7B; frontier-scale effects are inferred from observational data only.
- **Training regime**: SFT injection approximates but does not replicate pre-training contamination or RL post-training. The observational study provides preliminary RL evidence.
- **Unidimensional θ**: MMLU spans 57 subjects; a single θ per model oversimplifies multidimensional ability Reckase (2009); Burnell et al. (2023).
- **Coverage**: 63% at 95% CI means calibrated intervals are directional indicators, not precise credible intervals.
- **Model family**: Primary results on Qwen2.5; Llama-3.2-3B validates cross-family consistency (τ=0.87) but broader validation is needed.

8 Conclusion

ContamCal demonstrates that benchmark contamination effects are far more structured than "scores go up." Through controlled injection and IRT analysis across N=24 models spanning two families and four scales, we establish that contamination is benchmark-dependent (ARC forgetting vs. GSM8K inflation), scale-modulated (catastrophic to negligible), and domain-heterogeneous (Simpson's paradox within MMLU). Min-K++ directional reversal across 0.5B–3B and two model families constitutes a cautionary finding for the detection community. ContamCal provides conditional calibration when signal quality permits (ARC MAE=0.030) while honestly identifying when calibration should not be applied. The framework's primary value is not score correction but systematic characterization of contamination dynamics—revealing structure where previous work assumed uniformity.

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